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Flight path generation apparatus, aircraft system, flight path generation method, and readable medium

Abstract

A flight path generation apparatus (300) is configured to generate a flight path for an aircraft (100) capable of autonomously flying, and includes: an airframe information acquisition unit (301) configured to acquire airframe information including airframe IDs and take-off places of a plurality of aircraft (100); a movement information acquisition unit (302) configured to acquire movement information related to scheduled take-off times and destinations of the plurality of aircraft; a generation unit (303) configured to generate flight paths from the take-off places to landing places corresponding to the destinations based on the airframe information and the movement information; and a communication unit (304) configured to transmit the flight paths to the aircraft.

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Background/Summary

(1) This application is a National Stage Entry of PCT/JP2020/016767 filed on Apr. 16, 2020, the contents of all of which are incorporated herein by reference, in their entirety.

TECHNICAL FIELD

(2) The present disclosure relates to a flight path generation apparatus, an aircraft system, a flight path generation method, and a readable medium.

BACKGROUND ART

(3) Patent Literature 1 discloses a system for enabling a UAV (Unmanned Aerial Vehicle) to make a planned landing and an emergency landing. The UAV takes off from a transport vehicle and lands on the transport vehicle. In Patent Literature 1, a fixed fee may be paid to allow the UAV to land the transport vehicle, or a fee may be paid each time the UAV is allowed to land the transport vehicle.

(4) Patent Literature 2 discloses a management apparatus for a hangar facility in which an aircraft lands. The hangar facility management apparatus associates aircraft identification information, a landing time, and a port ID with each other, and transmits them to another hangar facility management apparatus.

CITATION LIST

Patent Literature

(5) Patent Literature 1: Published Japanese Translation of PCT International Publication for Patent Application, No. 2018-513456 Patent Literature 2: International Patent Publication No. WO2019/188250

SUMMARY OF INVENTION

Technical Problem

(6) There has been a demand for a technology for managing aircraft such as unmanned aerial vehicles more efficiently.

(7) The present disclosure has been made in order to solve the above-described problem, and an object thereof is to provide a flight path generation apparatus, an aircraft system, a flight path generation method, and a readable medium capable of efficiently managing a plurality of aircraft.

Solution to Problem

(8) A flight path generation apparatus according to the present disclosure is a flight path generation apparatus configured to generate a flight path for an aircraft capable of autonomously flying, including: an airframe information acquisition unit configured to acquire airframe information including airframe IDs and take-off places of a plurality of aircraft; a movement information acquisition unit configured to acquire movement information related to scheduled take-off times and destinations of the plurality of aircraft; a generation unit configured to generate flight paths from the take-off places to landing places corresponding to the destinations based on the airframe information and the movement information; and a communication unit configured to transmit the flight paths to the aircraft.

(9) An aircraft system according to the present disclosure is a flight path generation system configured to generate a flight path for an aircraft capable of autonomously flying, including: an airframe information acquisition unit configured to acquire airframe information including airframe IDs and take-off places of a plurality of aircraft; a movement information acquisition unit configured to acquire movement information related to scheduled take-off times and destinations of

the plurality of aircraft; a generation unit configured to generate flight paths from the take-off places to landing places corresponding to the destinations based on the airframe information and the movement information; a communication unit configured to transmit the flight paths to the aircraft; and an aircraft configured to autonomously fly along a flight path received from the communication unit.

(10) A flight path generation method according to the present disclosure is a flight path generation method for generating a flight path for an aircraft capable of autonomously flying, including: a step of acquiring airframe information including airframe IDs and take-off places of a plurality of aircraft; a step of acquiring movement information related to scheduled take-off times and destinations of the plurality of aircraft; a step of generating flight paths from the take-off places to landing places corresponding to the destinations based on the airframe information and the movement information; and a step of transmitting the flight paths to the aircraft.

(11) A readable medium according to the present disclosure is a non-transitory computer readable medium storing a program for causing a computer to perform a flight path generation method for generating a flight path for an aircraft capable of autonomously flying, the flight path generation method including: a step of acquiring airframe information including airframe IDs and take-off places of a plurality of aircraft; a step of acquiring movement information related to scheduled take-off times and destinations of the plurality of aircraft; a step of generating flight paths from the take-off places to landing places corresponding to the destinations based on the airframe information and the movement information; and a step of transmitting the flight paths to the aircraft.

Advantageous Effects of Invention

(12) According to the present disclosure, it is possible to provide a flight path generation apparatus, an aircraft system, a flight path generation method, and a readable medium capable of efficiently managing a plurality of aircraft.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a schematic view showing an aircraft system **1** according to a first example embodiment;

(2) FIG. 2 is a control block diagram of the aircraft system **1**;

(3) FIG. 3 is a schematic view showing a flight path generation apparatus according to another example embodiment; and

(4) FIG. 4 is a block diagram showing an example of a hardware configuration.

EXAMPLE EMBODIMENT

First Example Embodiment

(5) An aircraft system **1** according to a first example embodiment will be described with reference to FIG. 1. FIG. 1 schematically shows the aircraft system **1**. The aircraft system **1** includes a plurality of aircraft **100** and a flight path generation apparatus **300**.

(6) Each of the aircraft **100** is a rotary-wing aircraft including rotor blades **101**. As the rotor blades **101** are rotationally driven, lift and thrust are generated. Note that, although the aircraft **100** includes four rotor blades **101** in FIG. 1, the number of rotor blades is not limited to any particular number. The plurality of aircraft **100** may be the same type of aircraft (e.g., the same model), or may be different types of aircraft.

(7) Each of the aircraft **100** can autonomously fly. Examples of the aircraft **100** include drones, unmanned aerial vehicles (UAVs: Unmanned Aerial Vehicles), and flying cars. The aircraft **100** may be a vertical take-off and landing (Vtol: Vertical Take-Off and Landing) aircraft. The aircraft **100** may be a tilt-rotor aircraft. The aircraft **100** may be a helicopter. The aircraft **100** may be an

unmanned aircraft loaded with luggage or the like, or may be a manned aircraft in which a crew(s) and/or a passenger(s) are on board.

(8) The aircraft **100** can autonomously fly along a flight path (i.e., a flight route) from a take-off place to a landing place. For example, the aircraft **100** takes off from a take-off and landing facility and flies along a flight path. When the aircraft **100** has flown to the landing place corresponding to the destination, it lands on the landing place. The flight path is a 3D (three-dimensional) path from the take-off place to the landing place. The take-off place and the landing place are take-off and landing facilities designated in advance. Note that the take-off place and the landing place may be arbitrary places as long as there is an enough space for the landing therein. Needless to say, the take-off and landing facility on which the aircraft lands may be the same as the take-off and landing facility from which the aircraft has taken off.

(9) The flight path generation apparatus **300** generates flight paths for the aircraft **100**, which can autonomously fly. Then, the flight path generation apparatus **300** transmits the flight paths to the respective aircraft **100**. The flight path generation apparatus **300** is an information processing apparatus such as a computer. For example, the flight path generation apparatus **300** is a server apparatus connected to a network such as the Internet. The flight path generation apparatus **300** is not limited to a physically single apparatus. For example, a plurality of processors may perform later-described processes in a cooperative manner. The flight path generation apparatus **300** includes an airframe information acquisition unit **301**, a movement information acquisition unit **302**, a generation unit **303**, a communication unit **304**, and an environmental information acquisition unit **305**.

(10) The airframe information acquisition unit **301** acquires airframe information including airframe IDs and take-off places of a plurality of aircraft. The aircraft IDs are identification numbers or the like for identifying the respective aircraft. A unique airframe ID is assigned to each of the aircraft **100**. The take-off place is information indicating a place (coordinates) at which a take-off and landing facility in which the aircraft **100** is present is located. For example, the current position of the aircraft **100** is the take-off place.

(11) The airframe information may include performance information about the performance of the aircraft **100**. The performance information includes data on the weight, size, possible longest flight time, ability as to turning movement, tolerance to wind, maximum flight speed, and maximum flight altitude of the aircraft **100**. The performance information may include data on the remaining amount of battery and/or the remaining amount of fuel during flight. Further, the performance information may include information indicating whether the aircraft is a manned aircraft or an unmanned aircraft. The airframe information may include information indicating whether the aircraft is an emergency aircraft such as a police aircraft, a fire aircraft, or an ambulance aircraft. The airframe information acquisition unit **301** may specify the performance information of the aircraft **100** based on the type (e.g., the model) and the airframe ID thereof.

(12) The movement information acquisition unit **302** acquires movement information related to scheduled take-off times (scheduled take-off times) and destinations of the plurality of aircraft. The scheduled take-off time may be the current time or a time registered in advance. The estimated take-off times and the destinations are information entered by the users of the aircraft **100**. Note that the destination may be a name of a place, a name of a facility, an address, coordinates (a latitude and a longitude), or the like. Further, the destination may be an ID or the like of the take-off and landing facility itself. Further, the movement information may include an intermediate point(s) between the take-off place to the landing place (i.e., a place(s) that the aircraft passes through or stops in the flight from the take-off place to the landing place).

(13) The generation unit **303** generates a flight path (i.e., a flight route) based on the airframe information and the movement information. The flight path is a traveling path (i.e., a traveling route) from the take-off place to the landing place corresponding to the destination. The flight path is information indicating a trajectory that passes through target points of the aircraft **100**. Further, a

scheduled flight time may be associated with each of the target points in the flight path. The flight path is, for example, composed of a set of 3D (three-dimensional) coordinates indicating target points. Specifically, the flight path is data in which 3D coordinates are arranged in a chronological order. A flight path is generated by connecting 3D coordinates from one to another.

(14) The generation unit **303** generates, for example, congestion information based on the airframe information and the movement information. The congestion information is an index indicating the congestion level of aircraft **100**. The generation unit **303** divides the flight area into a plurality of spaces, and calculates, for each space, the number of aircraft that are simultaneously present in that space as congestion information. The generation unit **303** generates a flight path so that the number of aircraft present in any of the spaces does not exceed a predetermined number. Alternatively, the congestion information may be a value that is determined according to the distance between aircraft. The generation unit **303** generates a flight path so that a distance between one aircraft **100** and another aircraft does not decrease to a certain distance or shorter.

(15) The landing place indicates the place (coordinates) of the take-off and landing facility. The generation unit **303** generates a flight path from the take-off place to a take-off and landing facility near the destination. For example, in the case where the destination is a name of a place, a name of a facility, an address or the like, a take-off and landing facility closest to the destination can be used as the landing place. Needless to say, the user may enter the take-off and landing facility itself as the destination. The flight path is a path in a 3D space. The generation unit **303** generates a flight path for each of the aircraft **100**. The generation unit **303** generates a flight path when a destination, a scheduled flight time, and the like are input. For example, the generation unit **303** successively generates flight paths for the respective aircraft **100** in the order in which the movement information of these aircraft **100** have been input. Further, the generation unit **303** may generate a plurality of flight paths for one aircraft **100**. For example, when there are a plurality of take-off and landing facilities near the destination, the generation unit **303** may generate a flight path to each of the take-off and landing facilities.

(16) The environmental information acquisition unit **305** acquires environmental information about a flight environment. The environmental information may include, for example, weather information about the weather of the flight area. The weather information may include, for example, rain, fine weather, a wind speed, a wind direction, a precipitation, and the like. The environmental information acquisition unit **305** may divide the flight area into a plurality of areas, and acquire weather information for each of the divided areas.

(17) The environmental information may include information indicating the presence/absence of a suspicious aircraft. Examples of the suspicious aircraft include an aircraft of which the airframe ID is not registered in advance. Further, the environmental information may include information indicating the presence/absence of an emergency aircraft. For example, in the flight area, information indicating a place where a suspicious aircraft or an emergency aircraft flies and/or a time of the flight can be used as environmental information. Various sensors for detecting environmental information may be provided. For example, environmental information may be measured by a wind velocity sensor and/or a rain sensor. Further, various sensors may be provided in the aircraft **100**, so that environmental information may be acquired from the aircraft **100** at all times or as required.

(18) The communication unit **304** transmits flight paths to aircraft **100**. The communication unit **304** transmits a flight path to each of the aircraft **100**. The communication unit **304** performs processing, for example, in conformity with communication standards such as 5G, 4G, Wi-Fi (Registered Trademark), and Bluetooth (Registered Trademark). The communication unit **304** transmits radio signals to aircraft **100** and a terminal(s) **400** (which will be described later). Further, the communication unit **304** receives radio signals from the aircraft **100** and the terminal(s) **400**. In this way, data and information can be transmitted and received between the user side such as the aircraft **100** and the server side such as the flight path generation apparatus **300**.

(19) Next, configurations of the aircraft **100** and the terminal **400** will be described. FIG. 2 is a functional block diagram showing configurations of the aircraft **100** and the terminal **400**. The terminal **400** is an apparatus through which the user or the like of the aircraft **100** enters necessary information. The terminal **400** is, for example, an information processing apparatus such as a smartphone or a personal computer. The terminal **400** includes an input unit **401**, a display unit **402**, a terminal-side communication unit **403**, and a terminal control unit **404**. Note that at least some of the processes performed by the terminal **400** may be performed by a processor or the like provided in the aircraft **100**. Further, the terminal **400** may be disposed in the aircraft **100**.

(20) The input unit **401** includes an input device such as a touch panel, a keyboard, a mouse, and a voice input microphone, and receives inputs from the user. For example, the user enters a destination, a scheduled take-off time, and the like by operating the input unit **401**. Further, the user may enter an intermediate point(s) or the like between the take-off place and the landing place by operating the input unit **401**. The display unit **402** includes a display device and displays a window or the like through which the user enters instructions or the like.

(21) The terminal-side communication unit **403** wirelessly communicates with the aircraft **100** and the flight path generation apparatus **300**. In this way, movement information such as a destination and a scheduled take-off time is transmitted to the aircraft **100** or to the flight path generation apparatus **300**. The terminal control unit **404** controls each component. For example, the terminal control unit **404** includes a processor and a memory. The terminal control unit **404** controls each block by executing a control program stored in the memory.

(22) The aircraft **100** includes a flight control unit **111**, a driving mechanism **112**, an airframe-side communication unit **113**, an airframe-side sensor(s) **114**, and a battery **116**.

(23) The flight control unit **111** controls each component. For example, the driving mechanism **112** includes rotor blades **101** and a motor therefor, and generates lift and thrust necessary for flight. The flight control unit **111** outputs a drive signal for controlling the driving mechanism **112**. In the example shown in FIG. 1, the flight control unit **111** controls the driving mechanism **112** so that the driving mechanism **112** independently drives four rotor blades **101**. The flight control unit **111** controls the driving mechanism **112** so that the aircraft **100** autonomously flies to the landing place. The aircraft **100** flies along a flight path.

(24) The flight control unit **111** stores a flight path received from the flight path generation apparatus **300** in a memory or the like. The flight control unit **111** controls the driving mechanism **112** so that the aircraft **100** flies along the flight path received from the flight path generation apparatus **300**. For example, the flight control unit **111** controls the driving mechanism **112** so that the position of the own aircraft moves along the flight path. When the position of the own aircraft is deviated from the flight path due to wind or the like, the aircraft **100** flies so as to get close to the flight path. The position of the own aircraft can be detected by the airframe-side sensor **114**.

(25) The airframe-side communication unit **113** wirelessly communicates with the ground side, i.e., with the flight path generation apparatus **300** or the terminal **400**. The airframe-side communication unit **113** performs processing, for example, in conformity with communication standards such as 5G, 4G, Wi-Fi (Registered Trademark), and Bluetooth (Registered Trademark). The airframe-side communication unit **113** transmits radio signals to the ground side. The airframe-side communication unit **113** receives radio signals from the ground side. In this way, data and information can be transmitted and received between the aircraft **100** and the ground side. The communication between the flight path generation apparatus **300** and the terminal **400** may be wire communication. The airframe-side communication unit **113** receives a flight path from the flight path generation apparatus **300**.

(26) The airframe-side sensor **114** detects information about the flight state of the aircraft **100**. The airframe-side sensor **114** includes, for example, a gyro sensor that detects the attitude of the airframe. Further, the airframe-side sensor **114** may include a position sensor that detects the position of the own aircraft. As the position sensor, for example, a satellite positioning sensor such

as a GPS may be used. Further, the airframe-side sensor **114** may detect obstacles and other aircraft near the own aircraft. The flight control unit **111** controls the driving mechanism **112** based on the result of the detection by the airframe-side sensor **114**. In this way, the aircraft **100** can autonomously fly along the flight path from the take-off place to the landing place. The airframe-side sensor **114** is not limited to one sensor, and may include a plurality of sensors. The battery **116** supplies electric power to each apparatus (i.e., to each unit).

(27) The aircraft **100** is autonomously flying along the flight path. The flight path generation apparatus **300** generates, based on movement information and aircraft-type information of each of a plurality of aircraft **100**, flight paths for the respective aircraft **100**. In this way, it is possible to efficiently manage the plurality of aircraft **100**. For example, aircraft **100** can fly while avoiding a crowded area. Aircraft **100** can fly without getting close to any of the other aircraft **100**. Since aircraft **100** are not affected by any of the other aircraft, each of the aircraft **100** can fly more safely and more efficiently. For example, an aircraft **100** does not need to perform a movement for avoiding other aircraft **100** according to the result of the detection by the airframe-side sensor **114**. It is possible to shorten flight times and reduce fuel consumption in the whole system.

(28) Further, the generation unit **303** may generate a flight path based on performance information. For example, the generation unit **303** generates a flight path so that the performance (i.e., conditions) indicated by the performance information is satisfied. The performance information is the weight, size, possible longest flight time, ability as to turning movement, tolerance to wind, maximum flight speed, and maximum flight altitude of the aircraft **100**. The performance information may include the current battery remaining amount and/or the current fuel remaining amount. For example, when the power source is an electric motor, the battery remaining amount is included in the performance information.

(29) Further, when the power source is an internal combustion engine, the fuel remaining amount such as the remaining amount of gasoline is included in the performance information. Alternatively, when fuel cells are used as the battery **116**, the fuel remaining amount such as the remaining amount of hydrogen is included in the performance information. When an internal combustion engine and an electric motor are used together as the power source, both the battery remaining amount and the fuel remaining amount may be included in the performance information. Further, the performance information may include information indicating whether the aircraft is a manned aircraft or an unmanned aircraft. For example, when the possible longest flight time is included as the performance information, a flight path is generated so as not to exceed the possible longest flight time. Specifically, the generation unit **303** shortens the flight distance for an aircraft **100** of which the possible longest flight time is short. That is, the generation unit **303** generates a flight path along which the aircraft **100** can fly without exceeding the possible longest flight time. Needless to say, the generation unit **303** can generate a flight path so as to satisfy performance (i.e., conditions) other than the possible longest flight time.

(30) Further, in the case where the aircraft **100** is an emergency aircraft, the generation unit **303** may preferentially generate a flight path for that aircraft **100**, i.e., for the emergency aircraft. For example, the generation unit **303** generates a flight path so that the aircraft **100**, which is an emergency aircraft, can arrive at the destination or the intermediate point more quickly.

(31) The generation unit **303** may generate a flight path based on environmental information. The environmental information may include, for example, weather information about the weather in the flight area. The weather information may include, for example, rain, fine weather, a wind speed, a wind direction, a precipitation, and the like. Further, the environmental information includes movement information of aircraft which are flying near the own aircraft. Further, the environmental information may include information indicating that a suspicious unknown aircraft is flying. For example, the generation unit **303** can generate a flight path so as to avoid a rainy area and/or a windy area. Further, the generation unit **303** may generate a flight path based on the environmental information and the performance information. It is assumed that, for example, the performance

information includes a tolerance to wind. It is assumed that a wind velocity on an area-by-area basis is included in the environmental information. The generation unit **303** generates a flight path so as to avoid an area(s) in which the wind velocity exceeds a flyable wind velocity indicated by the tolerance to wind. Aircraft can be operated more safely and more efficiently. As described above, the generation unit **303** can generate a flight path by referring to the environmental information and the performance information.

(32) The generation unit **303** may generate a flight path in a dynamic manner. For example, when the flight path generation apparatus **300** has acquired new information, it changes already-generated flight paths. As a result, the flight paths of aircraft **100** that are currently flying are updated. Then, the flight path generation apparatus **300** transmits the updated flight paths to the aircraft **100**, which are flying. The aircraft **100** receive the updated flight paths. In this way, the aircraft **100** fly along the newly updated flight paths. In this way, a plurality of aircraft **100** can be operated more efficiently.

(33) For example, when the flight path generation apparatus **300** has received movement information for a new aircraft **100**, it generates a flight path for the new aircraft **100**. The generation unit **303** changes already-generated flight paths based on this newly-generated flight path. Note that movement information for the other aircraft **100** is used as environmental information. For example, in the case where the new aircraft **100** is an emergency aircraft, the flight paths for aircraft **100** other than the emergency aircraft are updated so that this new aircraft **100** can arrive at the destination or the intermediate point earlier. That is, the generation unit **303** updates the flight paths for aircraft **100** that are flying near the flight path of the emergency aircraft. In this way, aircraft can be operated more efficiently.

(34) Further, when information indicating that a suspicious aircraft such as a suspicious drone is flying is acquired as environmental information, the generation unit **303** updates flight paths. For example, the generation unit **303** updates flight paths so as to avoid an area(s) in which the suspicious aircraft is present (i.e., is flying).

(35) Alternatively, the generation unit **303** may change already-generated flight paths when the weather information has changed. For example, when bad weather such as heavy rain or strong wind occurs, the generation unit **303** updates flight paths so as to avoid the bad weather area(s). Alternatively, when the weather has improved, the area(s) in which aircraft have not been able to fly until then becomes a flyable area(s). In this case, the generation unit **303** updates flight paths so as to pass through the new flyable area. In this way, aircraft can be operated more efficiently. In this case, the generation unit **303** can update flight paths so as to satisfy the performance (i.e., conditions) such as a tolerance to wind.

(36) Further, when the weather has worsened and a strong wind occurs, the battery remaining amount or the fuel remaining amount may decrease more rapidly than expected. Alternatively, owing to changes in the weather, the battery remaining amount or the fuel remaining amount may decrease more slowly than expected. In such a case, the generation unit **303** may update flight paths based on the performance information indicating the latest battery remaining amount or the latest fuel remaining amount. That is, when the flight path generation apparatus **300** has newly acquired the battery remaining amount or the fuel remaining amount, the generation unit **303** may update flight paths.

(37) Further, the flight path generation apparatus **300** may charge the users of aircraft **100** some fees. For example, the generation unit **303** generates a flight path while associating a usage fee with it. Then, when the aircraft **100** has flown along the flight path, the user of the aircraft **100** is charged the usage fee. The usage fee may be paid in actual money (currency) or in reward points.

(38) When users are charged usage fees, a usage fee may be set for each landing facility. For example, a high price may be set for a take-off and landing facility which is frequently used, and a low price may be set for a take-off and landing facility which is infrequently used. Further, a usage fee may be set for each flight area. For example, a high price may be set for an area which is

frequently used, and a low price may be set for an area which is infrequently used. Further, the fee may be changed according to the time of day. For example, a high price may be set in a time period during which a lot of aircraft fly, and a low price may be set in a time period during which not many aircraft fly.

(39) As described above, the generation unit **303** can set a usage fee for each flight path according to the usefulness thereof. Needless to say, some flight paths may be free. Further, the generation unit **303** may adopt dynamic pricing for the usage fee. The generation unit **303** can set the usage fee according to the current congestion level or the like.

(40) Further, users may be charged monthly or annual usage fees. In this case, it is possible to set a plurality of fixed-price plans. In a low-price fixed-price plan, one usage fee is set as a normal usage fee. In a high-price fixed-price plan, one usage fee is reduced from a normal usage fee or is free.

(41) Further, the generation unit **303** may generate a plurality of flight paths for one aircraft **100**. The communication unit **304** transmits a plurality of flight paths to the terminal **400** or the aircraft **100** as flight path candidates. The flight path generation apparatus **300** presents (i.e., shows) the plurality of flight path candidates to the user. The display unit **402** of the terminal **400** displays the flight path candidates. The display unit **402** may present flight paths to the user through AR (Augmented Reality) display or VR (Virtual Reality) display. For example, the display unit **402** displays flight paths on a head-mounted display or the like. Alternatively, the display unit **402** may perform AR display by projecting a display image including flight paths onto the windshield or the like. Then, the user selects one of a plurality of displayed flight path candidates by operating the input unit **401**. The flight path candidate selected by the user is stored as the flight path in the memory in the aircraft **100**.

(42) Note that the flight path generation apparatus **300** may present (i.e., show) usage fees together with the flight path candidates. The generation unit **303** generates a plurality of flight paths while associating usage fees with the respective flight paths. The communication unit **304** transmits the usage fees together with the flight path candidates to the terminal **400**. Then, the display unit **402** displays the flight path candidates together with the usage fees.

(43) The user selects a flight path candidate according to the usage fee. For example, the user can select a flight path candidate of which the price is low though the flight time is long. Alternatively, the user can select a flight path candidate of which the flight time is short though the price is high. As described above, the user selects a suitable flight path from the presented flight path candidates. The user can select a suitable flight path, so that the usefulness can be improved.

(44) The generation unit **303** may generate a flight path based on the history of flights of an individual user, or based on preference information of the user which is input (i.e., registered) in advance. Further, when the aircraft **100** includes a camera as the airframe-side sensor **114**, the generation unit **303** can generate a flight path that passes through a scenic intermediate point(s). The user can view a video image of scenery taken (i.e., filmed) by (the camera of) the aircraft **100** during flight.

(45) When the user is charged the usage fee, it may be determined whether or not the aircraft **100** has flown along the flight path. The aircraft system **1** may determine whether or not to charge the user the usage fee according to the result of the determination. For example, when the aircraft **100** has flown along the flight path, the flight path generation apparatus **300** charges the user the usage fee. When the aircraft **100** has not flown along the flight path, the flight path generation apparatus **300** does not charge the user the usage fee. That is, when the aircraft **100** has flown along a path other than the presented flight path, the flight path generation apparatus **300** does not charge the user the usage fee.

(46) The flight path generation apparatus **300** may determine whether or not the aircraft **100** has flown along the flight path. The flight path generation apparatus **300** compares flight points through which the aircraft **100** has actually flown with the flight path, and thereby determines whether or not the aircraft has flown along the flight path. Then, the flight path generation

apparatus **300** can charge the user the fee according to the result of the comparison. That is, when the actual flight points are located along the selected flight path, the aircraft system **1** charges the user the usage fee.

(47) Note that the flight points of the aircraft **100** may be detected so that they are used as the basis (or the grounds) for the charging. The flight points of the aircraft **100** during the flight can be detected, for example, by using the ADS-B (Automatic Dependent Surveillance-Broadcast). Alternatively, the flight points of the aircraft **100** during the flight may be detected by using a satellite positioning sensor such as a GPS. The aircraft **100** stores the flight points in a chronological order. Then, the aircraft **100** transmits the flight points to the flight path generation apparatus **300** while associating the respective flight times with them. The flight path generation apparatus **300** determines whether or not the aircraft has flown along the flight path based on the flight points which are recorded in the chronological order.

(48) Although the flight path generation apparatus **300** performs a determination process and the like for the charging in the above description, the aircraft **100** and the terminal **400** may perform at least a part of the determination process. That is, any of the aircraft **100**, the terminal **400**, and the flight path generation apparatus **300** may perform the process for comparing the flight points with the flight path.

(49) Specifically, the aircraft **100** may store the trajectory of the flight points in a memory or the like, and the aircraft **100** may compare the flight points with the flight path. Alternatively, the aircraft **100** may transmit the flight points to the flight path generation apparatus **300** or the terminal **400** during the flight or after the landing. Then, the flight path generation apparatus **300** or the terminal **400** may compare the flight points with the flight path. Further, the aircraft system **1** charges the user the usage fee according to the result of the comparison.

Other Example Embodiment

(50) A flight path generation apparatus **300** according to another example embodiment will be described with reference to FIG. 3. The flight path generation apparatus **300** is a flight path generation apparatus that generates flight paths for aircraft that can autonomously fly. The flight path generation apparatus **300** includes an airframe information acquisition unit **301**, a movement information acquisition unit **302**, a generation unit **303**, and a communication unit **304**.

(51) The airframe information acquisition unit **301** acquires airframe information including airframe IDs and take-off places of a plurality of aircraft. The movement information acquisition unit **302** acquires movement information related to scheduled take-off times and destinations of a plurality of aircraft. The generation unit **303** generates flight paths from take-off places to landing places corresponding to the destinations based on the airframe information and the movement information. The communication unit **304** transmits the flight paths to the aircraft. In this way, aircraft can be efficiently operated.

(52) <Example of Hardware Configuration>

(53) An example of a hardware configuration of an apparatus according to the first example embodiment or other example embodiments will be described. FIG. 4 is a block diagram showing an example of a hardware configuration for performing information processing in each of the flight path generation apparatus **300**, the terminal **400**, and the aircraft **100**. Referring to FIG. 4, the flight path generation apparatus **300** or the like includes a network interface **601**, a processor **602**, and a memory **603**. The network interface **601** is used to communicate with a network node (e.g., an eNB, an MME, or a P-GW). The network interface **601** may include, for example, a network interface card (NIC) in conformity with IEEE 802.3 series. Note that the eNB represents an evolved Node B, the MME represents a Mobility Management Entity, and the P-GW represents a Packet Data Network Gateway. The IEEE stands for Institute of Electrical and Electronics.

(54) The processor **602** loads software (a computer program) from the memory **603** and executes the loaded software, so that the processor **602** performs the processes performed by the flight path generation apparatus **300** or the like described in the above-described example embodiment. The

processor **602** may be, for example, a microprocessor, an MPU, or a CPU. The processor **602** may include a plurality of processors.

(55) The memory **603** is composed of a combination of a volatile memory and a nonvolatile memory. The memory **603** may include a storage remotely located from the processor **602**. In this case, the processor **602** may access the memory **603** through an I/O (Input/Output) interface (not shown).

(56) In the example shown in FIG. **4**, the memory **603** is used to store a group of software modules. The processor **602** loads the group of software modules (a computer program(s)) from the memory **603** and executes the loaded software modules, so that the processor **602** can perform the processes performed by the flight path generation apparatus **300** or the like described in the above-described example embodiment.

(57) As described above with reference to FIG. **4**, each of the processors included in the flight path generation apparatus **300** and the like in the above-described example embodiment executes one or a plurality of programs including a set of instructions for causing a computer to perform the algorithm described above with reference to the drawings.

(58) In the above-described examples, the program can be stored and provided to a computer using any type of non-transitory computer readable media. Non-transitory computer readable media include any type of tangible storage media. Examples of non-transitory computer readable media include magnetic storage media (floppy disks, magnetic tapes, hard disk drives), optical magnetic storage media (e.g., magneto-optical disks), CD-ROM (Read Only Memory), CD-R, CD-R/W, and semiconductor memories (e.g., mask ROM, PROM (Programmable ROM), EPROM (Erasable PROM), flash ROM, and RAM (Random Access Memory)). Further, the program may be provided to a computer using any type of transitory computer readable media. Examples of transitory computer readable media include electric signals, optical signals, and electromagnetic waves. Transitory computer readable media can provide the program to a computer via a wired communication line (e.g., electric wires, and optical fibers) or a wireless communication line.

(59) Note that the present disclosure is not limited to the above-described example embodiments, and they may be modified as appropriate without departing from the scope and spirit of the disclosure.

(60) Although the present invention has been described with reference to example embodiments (and examples), the present invention is not limited to the above-described example embodiments (and examples). The configuration and details of the present invention may be modified within the scope of the present invention in various ways that can be understood by those skilled in the art.

(61) The whole or part of the example embodiments disclosed above can be described as, but not limited to, the following supplementary notes.

(62) (Supplementary Note 1)

(63) A flight path generation apparatus configured to generate a flight path for an aircraft capable of autonomously flying, comprising: an airframe information acquisition unit configured to acquire airframe information including airframe IDs and take-off places of a plurality of aircraft; a movement information acquisition unit configured to acquire movement information related to scheduled take-off times and destinations of the plurality of aircraft; a generation unit configured to generate flight paths from the take-off places to landing places corresponding to the destinations based on the airframe information and the movement information; and a communication unit configured to transmit the flight paths to the aircraft.

(Supplementary Note 2)

(64) The flight path generation apparatus described in Supplementary note 1, wherein the generation unit generates a plurality of flight paths for one aircraft, and the communication unit transmits the plurality of flight paths as flight path candidates.

(Supplementary Note 3)

(65) The flight path generation apparatus described in Supplementary note 2, wherein the

generation unit generates a plurality of flight paths while associating a usage fee with each of the flight paths, and the communication unit transmits the usage fees together with the flight path candidates.

(Supplementary Note 4)

(66) The flight path generation apparatus described in Supplementary note 3, wherein a flight point through which the aircraft has actually flown is compared with the flight path, and the usage fee is charged.

(67) (Supplementary Note 5)

(68) The flight path generation apparatus described in any one of Supplementary notes 1 to 4, wherein the airframe information includes performance information about performance of the aircraft.

(69) (Supplementary Note 6)

(70) The flight path generation apparatus described in Supplementary note 5, wherein the flight path is updated based on the performance information.

(71) (Supplementary Note 7)

(72) The flight path generation apparatus described in any one of Supplementary notes 1 to 6, further comprising an environmental information acquisition unit configured to acquire environmental information about a flight environment, wherein the flight path is updated based on the environmental information.

(Supplementary Note 8)

(73) An aircraft system configured to generate a flight path for an aircraft capable of autonomously flying, comprising: an airframe information acquisition unit configured to acquire airframe information including airframe IDs and take-off places of a plurality of aircraft; a movement information acquisition unit configured to acquire movement information related to scheduled take-off times and destinations of the plurality of aircraft; a generation unit configured to generate flight paths from the take-off places to landing places corresponding to the destinations based on the airframe information and the movement information; a communication unit configured to transmit the flight paths to the aircraft; and an aircraft configured to autonomously fly along a flight path received from the communication unit.

(Supplementary Note 9)

(74) The aircraft system described in Supplementary note 8, further comprising an airframe-side sensor provided to detect a flight point, wherein the generation unit generates the flight path while associating a usage fee with the flight path, and a flight point through which the aircraft has flown is compared with the flight path, and a user is charged the user fee associated with the flight path according to a result of the comparison.

(Supplementary Note 10)

(75) The aircraft system described in Supplementary note 8, wherein the generation unit generates a plurality of flight paths for one aircraft, and the communication unit transmits the plurality of flight paths as flight path candidates.

(Supplementary Note 11)

(76) The aircraft system described in Supplementary note 10, wherein the generation unit generates a plurality of flight paths while associating a usage fee with each of the flight paths, and the communication unit transmits the usage fees together with the flight path candidates.

(Supplementary Note 12)

(77) The aircraft system described in Supplementary note 10, wherein a flight point through which the aircraft has actually flown is compared with the flight path, and the usage fee is charged.

(78) (Supplementary Note 13)

(79) The aircraft system described in any one of Supplementary notes 8 to 12, wherein the airframe information includes performance information about performance of the aircraft.

(80) (Supplementary Note 14)

(81) The aircraft system described in Supplementary note 13 wherein the flight path is updated based on the performance information.

(82) (Supplementary Note 15)

(83) The aircraft system described in any one of Supplementary notes 8 to 14, further comprising an environmental information acquisition unit configured to acquire environmental information about a flight environment, wherein the flight path is updated based on the environmental information.

(Supplementary Note 16)

(84) A flight path generation method for generating a flight path for an aircraft capable of autonomously flying, comprising: a step of acquiring airframe information including airframe IDs and take-off places of a plurality of aircraft; a step of acquiring movement information related to scheduled take-off times and destinations of the plurality of aircraft; a step of generating flight paths from the take-off places to landing places corresponding to the destinations based on the airframe information and the movement information; and a step of transmitting the flight paths to the aircraft.

(Supplementary Note 17)

(85) A non-transitory computer readable medium storing a program for causing a computer to perform a flight path generation method for generating a flight path for an aircraft capable of autonomously flying, the flight path generation method comprising: a step of acquiring airframe information including airframe IDs and take-off places of a plurality of aircraft; a step of acquiring movement information related to scheduled take-off times and destinations of the plurality of aircraft; a step of generating flight paths from the take-off places to landing places corresponding to the destinations based on the airframe information and the movement information; and a step of transmitting the flight paths to the aircraft.

REFERENCE SIGNS LIST

(86) **100** AIRCRAFT **101** ROTOR BLADE **111** FLIGHT CONTROL UNIT **112** DRIVING MECHANISM **113** AIRFRAME-SIDE COMMUNICATION UNIT **114** AIRFRAME-SIDE SENSOR **116** BATTERY **300** FLIGHT ROUTE GENERATION APPARATUS **301** AIRFRAME INFORMATION ACQUISITION UNIT **302** MOVEMENT INFORMATION ACQUISITION UNIT **303** GENERATION UNIT **304** COMMUNICATION UNIT **305** ENVIRONMENTAL INFORMATION ACQUISITION UNIT **400** TERMINAL **401** INPUT UNIT **402** DISPLAY UNIT **403** TERMINAL-SIDE COMMUNICATION UNIT **404** TERMINAL CONTROL UNIT

Claims

1. A flight path generation apparatus comprising: at least one memory configured to store instructions; and at least one processor configured to execute the instructions stored in the memory to: acquire airframe information including airframe IDs and take-off places of a plurality of aircrafts; acquire movement information related to scheduled take-off times and destinations of the plurality of aircrafts; generate, as flight path candidates, a plurality of flight paths from the take-off places to landing places corresponding to the destinations based on the airframe information and the movement information; generate congestion information indicating congestion level of the plurality of aircrafts the based on the airframe information and the movement information; generate the flight path candidates while associating a usage fee with each flight path candidate; set the usage fee according to the congestion information; transmit a plurality of usage fees together with the flight path candidates; select a flight path from among the flight path candidates for one aircraft, from among the plurality of aircrafts; and control the one aircraft to fly along the selected flight path.

2. The flight path generation apparatus according to claim 1, wherein a flight point through which the one aircraft has actually flown is compared with the selected flight path, and the usage fee is

charged.

3. The flight path generation apparatus according to claim 1, wherein the airframe information includes performance information about performance of the one aircraft.

4. The flight path generation apparatus according to claim 3, wherein the selected flight path is updated based on the performance information.

5. The flight path generation apparatus according to claim 1, further comprising an environmental information acquisition unit configured to acquire environmental information about a flight environment, wherein the selected flight path is updated based on the environmental information.

6. A flight path generation method for generating a flight path for an aircraft capable of autonomously flying, comprising: acquiring airframe information including airframe IDs and take-off places of a plurality of aircrafts; acquiring movement information related to scheduled take-off times and destinations of the plurality of aircrafts; generating, as flight path candidates, a plurality of flight paths from the take-off places to landing places corresponding to the destinations based on the airframe information and the movement information; generating congestion information indicating congestion level of the plurality of aircrafts based on the airframe information and the movement information; generating the flight path candidates while associating a usage fee with each flight path candidate; setting the usage fee according to the congestion information; transmitting a plurality of usage fees together with the flight path candidates; selecting a flight path from among the flight path candidates for one aircraft, from among the plurality of aircrafts; and controlling the one aircraft to fly along the selected flight path.

7. The flight path generation method according to claim 6, wherein a flight point through which the one aircraft has actually flown is compared with the selected flight path, and the usage fee is charged.

8. The flight path generation method according to claim 6, wherein the airframe information includes performance information about performance of the selected aircraft.

9. The flight path generation method according to claim 8, wherein the selected flight path is updated based on the performance information.

10. The flight path generation method according to claim 6, further comprising an environmental information acquisition unit configured to acquire environmental information about a flight environment, wherein the selected flight path is updated based on the environmental information.

11. A non-transitory computer readable medium storing a program for causing a computer to perform a flight path generation method for generating a flight path for an aircraft capable of autonomously flying, the flight path generation method comprising: acquiring airframe information including airframe IDs and take-off places of a plurality of aircrafts; acquiring movement information related to scheduled take-off times and destinations of the plurality of aircrafts; generating, as flight path candidates, a plurality of flight paths from the take-off places to landing places corresponding to the destinations based on the airframe information and the movement information; generating congestion information indicating congestion level of the plurality of aircrafts based on the airframe information and the movement information; generating the flight path candidates while associating a usage fee with each flight path candidate; setting the usage fee according to the congestion information; transmitting a plurality of usage fees together with the flight path candidates; selecting a flight path from among the flight path candidates for one aircraft, from among the plurality of aircrafts; and controlling the one aircraft to fly along the selected flight path.

12. The non-transitory computer readable medium according to claim 11, wherein a flight point through which the one aircraft has actually flown is compared with the selected flight path, and the usage fee is charged.

13. The non-transitory computer readable medium according to claim 11, wherein the airframe information includes performance information about performance of the one aircraft.

14. The non-transitory computer readable medium according to claim 13, wherein the selected flight path is updated based on the performance information.
